# Twelfth Grade Tile Task

Standard S.CP.1

Name: _____ Date: _____

12

**1. a)** Outcomes of a spinner with 1 equal sections

**b)** Outcomes of a spinner with 7 equal sections

**c)** Outcomes of a fair die?

**2. a)** Outcomes of flipping two coins?

**b)** Outcomes of a spinner with 5 equal sections

**c)** Outcomes of a spinner with 3 equal sections

**3. a)** Outcomes of a spinner with 8 equal sections

**b)** Outcomes of a spinner with 9 equal sections

**c)** Outcomes of a spinner with 2 equal sections

**Tile box: cut out the tiles and move them to your answer boxes**

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|

# Twelfth Grade Tile Task

Standard S.CP.2

Name: _____ Date: _____

12

**1.** **a)** Outcomes where two dice sum to 7? ☐

**b)** Outcomes where a die is 5? ☐

**c)** Outcomes where a die shows 7? ☐

**2.** **a)** Outcomes of a spinner with 9 equal sections? ☐

**b)** Outcomes of a spinner with 7 equal sections? ☐

**c)** Outcomes where two dice sum to 8? ☐

**3.** **a)** Outcomes where two dice sum to 4? ☐

**b)** Outcomes where a die is greater than 2? ☐

**c)** Outcomes of a spinner with 8 equal sections? ☐

**Tile box: cut out the tiles and move them to your answer boxes**

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|

# Twelfth Grade Tile Task

Standard S.CP.3

Name: _____ Date: _____

12

**1. a)** Favorable 1 out of 7 equally likely outcomes: numerator ☐

**b)** Favorable 8 out of 9 equally likely outcomes: numerator ☐

**c)** Favorable 4 out of 6 equally likely outcomes: numerator ☐

**2. a)** Favorable 3 out of 5 equally likely outcomes: numerator ☐

**b)** Favorable 2 out of 5 equally likely outcomes: numerator ☐

**c)** Favorable 7 out of 7 equally likely outcomes: numerator ☐

**3. a)** Favorable 9 out of 9 equally likely outcomes: numerator ☐

**b)** Favorable 6 out of 8 equally likely outcomes: numerator ☐

**c)** Favorable 5 out of 7 equally likely outcomes: numerator ☐

**Tile box: cut out the tiles and move them to your answer boxes**

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|

**TILE TASKS: TWELFTH GRADE**

# Twelfth Grade Tile Task

Standard S.CP.4

Name: _____ Date: _____

12

1. **a)** Two-way table: 5 yes and 0 no. Total? ☐
   **b)** Two-way table: 7 yes and 2 no. Total? ☐
   **c)** Two-way table: 7 yes and 1 no. Total? ☐

2. **a)** Two-way table: 1 yes and 1 no. Total? ☐
   **b)** Two-way table: 2 yes and 1 no. Total? ☐
   **c)** Two-way table: 4 yes and 3 no. Total? ☐

3. **a)** Two-way table: 0 yes and 1 no. Total? ☐
   **b)** Two-way table: 4 yes and 2 no. Total? ☐
   **c)** Two-way table: 0 yes and 4 no. Total? ☐

**Tile box: cut out the tiles and move them to your answer boxes**

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|

# Twelfth Grade Tile Task

Standard S.CP.5

Name: _____ Date: _____

12

**1. a)** 4 of 10 students are in both groups. How many? ☐

**b)** 3 of 10 students are in both groups. How many? ☐

**c)** 8 of 10 students are in both groups. How many? ☐

**2. a)** 1 of 10 students are in both groups. How many? ☐

**b)** 5 of 10 students are in both groups. How many? ☐

**c)** 0 of 10 students are in both groups. How many? ☐

**3. a)** 7 of 10 students are in both groups. How many? ☐

**b)** 2 of 10 students are in both groups. How many? ☐

**c)** 9 of 10 students are in both groups. How many? ☐

**Tile box: cut out the tiles and move them to your answer boxes**

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|

# Twelfth Grade Tile Task

Standard S.CP.6

Name: _____ Date: _____

12

**1. a)** Chance of an event is 7 out of 10. Numerator? ☐

**b)** Chance of an event is 0 out of 10. Numerator? ☐

**c)** Chance of an event is 6 out of 10. Numerator? ☐

**2. a)** Chance of an event is 9 out of 10. Numerator? ☐

**b)** Chance of an event is 1 out of 10. Numerator? ☐

**c)** Chance of an event is 5 out of 10. Numerator? ☐

**3. a)** Chance of an event is 8 out of 10. Numerator? ☐

**b)** Chance of an event is 3 out of 10. Numerator? ☐

**c)** Chance of an event is 4 out of 10. Numerator? ☐

**Tile box: cut out the tiles and move them to your answer boxes**

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|

# Twelfth Grade Tile Task

Standard S.CP.7

Name: _____ Date: _____

12

**1. a)** 4 of 10 have the trait. How many have it? ☐

**b)** 3 of 10 have the trait. How many have it? ☐

**c)** 5 of 10 have the trait. How many have it? ☐

**2. a)** 7 of 10 have the trait. How many have it? ☐

**b)** 2 of 10 have the trait. How many have it? ☐

**c)** 9 of 10 have the trait. How many have it? ☐

**3. a)** 6 of 10 have the trait. How many have it? ☐

**b)** 8 of 10 have the trait. How many have it? ☐

**c)** 0 of 10 have the trait. How many have it? ☐

**Tile box: cut out the tiles and move them to your answer boxes**

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|

# Twelfth Grade Tile Task

Standard S.IC.1

Name: _____ Date: _____

12

**1.** **a)** A random sample has 7 people. ☐

**b)** A random sample has 4 people. ☐

**c)** A random sample has 1 people. ☐

**2.** **a)** A random sample has 6 people. ☐

**b)** A random sample has 2 people. ☐

**c)** A random sample has 9 people. ☐

**3.** **a)** A random sample has 3 people. ☐

**b)** A random sample has 8 people. ☐

**c)** A random sample has 5 people. ☐

**Tile box: cut out the tiles and move them to your answer boxes**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |

# Twelfth Grade Tile Task

Standard S.IC.2

Name: _____ Date: _____

12

**1.** **a)** Simulation: heads in 0 of 10 trials. ☐

**b)** Simulation: heads in 6 of 10 trials. ☐

**c)** Simulation: heads in 9 of 10 trials. ☐

**2.** **a)** Simulation: heads in 5 of 10 trials. ☐

**b)** Simulation: heads in 8 of 10 trials. ☐

**c)** Simulation: heads in 3 of 10 trials. ☐

**3.** **a)** Simulation: heads in 7 of 10 trials. ☐

**b)** Simulation: heads in 1 of 10 trials. ☐

**c)** Simulation: heads in 4 of 10 trials. ☐

**Tile box: cut out the tiles and move them to your answer boxes**

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|

# Twelfth Grade Tile Task

Standard S.IC.3

Name: _____ Date: _____

12

1. **a)** Survey sample size of 6 people ☐
   **b)** Survey sample size of 5 people ☐
   **c)** Survey sample size of 3 people ☐

2. **a)** Survey sample size of 8 people ☐
   **b)** Survey sample size of 7 people ☐
   **c)** Survey sample size of 1 people ☐

3. **a)** Survey sample size of 4 people ☐
   **b)** Survey sample size of 9 people ☐
   **c)** Survey sample size of 2 people ☐

**Tile box: cut out the tiles and move them to your answer boxes**

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|

# Twelfth Grade Tile Task

Standard S.IC.4

Name: _____ Date: _____

12

**1. a)** In a sample of 10, 3 said yes. How many said yes? ☐

**b)** In a sample of 10, 7 said yes. How many said yes? ☐

**c)** In a sample of 10, 8 said yes. How many said yes? ☐

**2. a)** In a sample of 10, 1 said yes. How many said yes? ☐

**b)** In a sample of 10, 2 said yes. How many said yes? ☐

**c)** In a sample of 10, 0 said yes. How many said yes? ☐

**3. a)** In a sample of 10, 9 said yes. How many said yes? ☐

**b)** In a sample of 10, 6 said yes. How many said yes? ☐

**c)** In a sample of 10, 5 said yes. How many said yes? ☐

**Tile box: cut out the tiles and move them to your answer boxes**

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|

# Twelfth Grade Tile Task

Standard S.IC.5

Name: _____ Date: _____

12

**1. a)** Group A mean 7, group B mean 5: difference? ☐

**b)** Group A mean 9, group B mean 5: difference? ☐

**c)** Group A mean 6, group B mean 1: difference? ☐

**2. a)** Group A mean 4, group B mean 3: difference? ☐

**b)** Group A mean 9, group B mean 2: difference? ☐

**c)** Group A mean 9, group B mean 0: difference? ☐

**3. a)** Group A mean 8, group B mean 5: difference? ☐

**b)** Group A mean 8, group B mean 2: difference? ☐

**c)** Group A mean 8, group B mean 0: difference? ☐

**Tile box: cut out the tiles and move them to your answer boxes**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |

# Twelfth Grade Tile Task

Standard S.IC.6

Name: _____ Date: _____

12

**1. a)** Groups compared in a nine-group experiment? ☐

**b)** Groups compared in a four-group experiment? ☐

**c)** Groups compared in a three-group experiment? ☐

**2. a)** Groups compared in a five-group experiment? ☐

**b)** Groups compared in a six-group experiment? ☐

**c)** Groups compared in a one-group experiment? ☐

**3. a)** Groups compared in a seven-group experiment? ☐

**b)** Groups compared in a two-group experiment? ☐

**c)** Groups compared in an eight-group experiment? ☐

**Tile box: cut out the tiles and move them to your answer boxes**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |

# Twelfth Grade Tile Task

Standard S.ID.3

Name: _____ Date: _____

12

**1.** **a)** Median of 3, 6, 10 ☐

**b)** Median of 8, 6, 9 ☐

**c)** Median of 2, 8, 4 ☐

**2.** **a)** Median of 1, 10, 9 ☐

**b)** Median of 4, 10, 7 ☐

**c)** Median of 2, 6, 0 ☐

**3.** **a)** Median of 0, 6, 3 ☐

**b)** Median of 2, 5, 7 ☐

**c)** Median of 8, 0, 1 ☐

**Tile box: cut out the tiles and move them to your answer boxes**

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|

# Twelfth Grade Tile Task

Standard S.ID.4

Name: _____ Date: _____

**12**

**1. a)** Standard deviations in the 95% rule? ☐

**b)** Mean of a normal curve centered at 5? ☐

**c)** Mean of a normal curve centered at 9? ☐

**2. a)** Mean of a normal curve centered at 6? ☐

**b)** Standard deviations in the 68% rule? ☐

**c)** Mean of a normal curve centered at 7? ☐

**3. a)** Mean of a normal curve centered at 4? ☐

**b)** Mean of a normal curve centered at 8? ☐

**c)** Standard deviations in the 99.7% rule? ☐

**Tile box: cut out the tiles and move them to your answer boxes**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **0** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** |

# Twelfth Grade Tile Task

Standard S.MD.1

Name: _____ Date: _____

12

**1.** **a)** Expected value of a game that pays 8 with probability 1 ☐

**b)** Expected value of a game that pays 5 with probability 1 ☐

**c)** Expected value of a game that pays 1 with probability 1 ☐

**2.** **a)** Expected value of a game that pays 4 with probability 1 ☐

**b)** Expected value of a game that pays 3 with probability 1 ☐

**c)** Expected value of a game that pays 0 with probability 1 ☐

**3.** **a)** Expected value of a game that pays 6 with probability 1 ☐

**b)** Expected value of a game that pays 2 with probability 1 ☐

**c)** Expected value of a game that pays 7 with probability 1 ☐

**Tile box: cut out the tiles and move them to your answer boxes**

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|

# Twelfth Grade Tile Task

Standard S.MD.2

Name: _____ Date: _____

12

**1. a)** Weighted average of 9 (weight 1) and 7 (weight 1) ☐

**b)** Weighted average of 0 (weight 1) and 6 (weight 1) ☐

**c)** Weighted average of 1 (weight 1) and 1 (weight 1) ☐

**2. a)** Weighted average of 9 (weight 1) and 9 (weight 1) ☐

**b)** Weighted average of 7 (weight 1) and 1 (weight 1) ☐

**c)** Weighted average of 4 (weight 1) and 0 (weight 1) ☐

**3. a)** Weighted average of 6 (weight 1) and 8 (weight 1) ☐

**b)** Weighted average of 7 (weight 1) and 5 (weight 1) ☐

**c)** Weighted average of 7 (weight 1) and 3 (weight 1) ☐

**Tile box: cut out the tiles and move them to your answer boxes**

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|

# Twelfth Grade Tile Task

Standard S.MD.3

Name: _____ Date: _____

12

1. **a)** Outcomes of a spinner with 1 sections: count ☐
   **b)** Outcomes of a spinner with 8 sections: count ☐
   **c)** Outcomes of a spinner with 3 sections: count ☐

2. **a)** Outcomes of a spinner with 4 sections: count ☐
   **b)** Outcomes of a spinner with 6 sections: count ☐
   **c)** Outcomes of two coins with exactly one head ☐

3. **a)** Outcomes of a spinner with 7 sections: count ☐
   **b)** Outcomes of a spinner with 9 sections: count ☐
   **c)** Outcomes of a spinner with 5 sections: count ☐

**Tile box: cut out the tiles and move them to your answer boxes**

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|

# Twelfth Grade Tile Task

Standard S.MD.4

Name: _____ Date: _____

12

**1.** **a)** Expected payoff of a fair game paying 1 ☐

**b)** Expected payoff of a fair game paying 8 ☐

**c)** Expected payoff of a fair game paying 7 ☐

**2.** **a)** Expected payoff of a fair game paying 4 ☐

**b)** Expected payoff of a fair game paying 9 ☐

**c)** Expected payoff of a fair game paying 5 ☐

**3.** **a)** Expected payoff of a fair game paying 3 ☐

**b)** Expected payoff of a fair game paying 6 ☐

**c)** Expected payoff of a fair game paying 2 ☐

**Tile box: cut out the tiles and move them to your answer boxes**

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|

# Twelfth Grade Tile Task

Standard S.MD.6

Name: _____ Date: _____

**12**

**1. a)** Fair price of a game paying 9 with probability 1 ☐

**b)** Fair price of a game paying 6 with probability 1 ☐

**c)** Fair price of a game paying 2 with probability 1 ☐

**2. a)** Fair price of a game paying 7 with probability 1 ☐

**b)** Fair price of a game paying 8 with probability 1 ☐

**c)** Fair price of a game paying 0 with probability 1 ☐

**3. a)** Fair price of a game paying 4 with probability 1 ☐

**b)** Fair price of a game paying 5 with probability 1 ☐

**c)** Fair price of a game paying 3 with probability 1 ☐

**Tile box: cut out the tiles and move them to your answer boxes**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **0** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** |